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Addressing micronutrient deficiencies in rainfed Vertisols: Integrated nutrient management for improving soybean-chickpea productivity and protein security

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ABSTRACT

Micronutrient deficiencies and depleting soil organic matter limit the productivity of soybean-chickpea cropping systems in rainfed Vertisols, yet the optimal combination of application methods for direct and residual nutrient benefits remains unestablished. A three-year field experiment (2016-17 to 2018-19) was conducted in the Vertisols of Central India to evaluate farmyard manure and micronutrient management strategies combining zinc and molybdenum applied through soil, seed treatment, and foliar methods on soybean, with their residual effects assessed on the succeeding chickpea. FYM at 5 t ha^{-1} increased soybean yield by 25.8% and produced a significant residual effect on chickpea, with 19.2% higher yield and 45.5% greater nodule dry weight. Among the micronutrient strategies, soil application of $ZnSO_4$ @ 25 kg ha^{-1} and ammonium molybdate @ 1 kg ha^{-1} achieved the highest system productivity (4464 kg ha^{-1} soybean equivalent yield), net returns (₹94,332 ha^{-1}), and benefit-cost ratio (3.52), representing increases of 32.7%, 54.7%, and 53.0% over the control, respectively. The application method optimal for soybean (ammonium molybdate @ 1 g kg^{-1} seed treatment + foliar application of $ZnSO_4$ @ 0.5%) differed from that providing the greatest residual benefit to chickpea (soil-applied Zn + Mo), demonstrating that system-level optimisation requires evaluation across both crops. The system protein productivity marked 1081 kg ha^{-1}, equivalent to a protein supply capacity of 51 persons ha^{-1} year^{-1}. Structural equation modelling identified yield characters as the decisive yield determinants in both crops. These findings demonstrate that integrated soil application of Zn and Mo with FYM can enhance system productivity, profitability, and protein security in legume-based cropping systems in Vertisols of Central India.

1. Introduction

India has the world's largest area under rainfed agriculture, with approximately 86 million hectares, contributing 85% of pulse and 72% of oilseed production [1,2]. These systems are important for food and protein security [3], yet remain constrained by erratic rainfall, declining soil fertility and widespread micronutrient deficiencies [4,5].

Legume-based cropping systems offer advantages through biological nitrogen fixation, high seed protein, and tolerance to abiotic stress [6,7]. Soybean (*Glycine* max L.) is the leading kharif oilseed crop, with Madhya Pradesh, the "Soya State" of India, contributing approximately 59% of the national acreage [8]. Following soybean, chickpea (*Cicer arietinum* L.) is commonly grown in the residual moisture of the monsoon season [9]. This soybean-chickpea rotation is the second most important cropping system after soybean-wheat in the Vertisols of Central India [10], attractive to resource-constrained smallholder farmers due to low capital requirements and carryover benefits from inputs applied to soybean [11]. Despite these advantages, protein supply from these systems remains below its potential. Micronutrient deficiencies in the production system contribute to reduced crop productivity [5] and dietary nutrient gaps in the population [12,13]. These deficiencies can be addressed through integrated nutrient management (INM), combining

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with organic amendments, inorganic micronutrients and biofertilizers [14].

Application of farmyard manure (FYM) improves soil organic carbon, nutrient-supplying capacity and physical properties of Vertisols, which suffer progressive organic matter decline under high temperatures and intensive cropping [15–17]. Long-term studies on these soils have shown that FYM combined with inorganic fertilisers sustains soil health and increases crop yields [18]. Synergistic effects of FYM with micronutrients on growth and yield have also been observed in soybean-chickpea sequences [19,20]. However, these evaluations have predominantly measured single-crop responses rather than the simultaneous direct and residual effects within one rotation cycle.

Zinc (Zn) deficiency is widespread in Indian rainfed agriculture, affecting approximately 48% of soils nationally and 66.9% in Madhya Pradesh [21,22], and is projected to rise further [23]. A national assessment identified Zn as a limiting micronutrient in the soybean-growing Vertisols of Central India [24,25]. In legumes, Zn is essential for reproductive development, nodulation and seed quality [26–28]. Field studies have reported chickpea yield increases of 20-31% with Zn application [29,30]. For soybean in Central India, soil application of $ZnSO_4$ at 25 kg ha^{-1} or foliar spray of 0.5% $ZnSO_4$ at 30 and 45 days after sowing are recommended management practices [24]. In field trials, basal application of 25 kg $ZnSO_4$ ha^{-1} along with recommended fertilisers improved soybean grain yield by 37-86% over the unfertilised control [24,31]. However, whether the application method (soil, seed treatment, or foliar spray) optimal for the directly-fertilised soybean also provides the best residual benefit to the succeeding chickpea remains to be established.

Molybdenum (Mo), a cofactor of nitrogenase and nitrate reductase [32–34], is critical for legume productivity. Its deficiency severely impairs both symbiotic nitrogen fixation and nitrate assimilation [35]. In the soybean-chickpea belt of Madhya Pradesh, soils under long-term legume rotations contain Mo below the critical limit of 0.2 ppm [36,37]. Soil application of ammonium molybdate at 0.5 kg ha^{-1} substantially improved soybean productivity in Central India [24,38]. In more severely Mo-depleted soils, ammonium molybdate at 1 kg ha^{-1} increased chickpea seed yield by up to 14-35% through soil application [36,39,40]. A lower-cost seed treatment at 1 g kg^{-1} seed was equally effective and economically remunerative [36]. Combined application of Zn and Mo with organic manure can produce synergistic effects on nodulation, nitrogen accumulation and protein content [41,42]. In this combination, Zn primarily supports reproductive partitioning while Mo drives biological nitrogen fixation.

Despite the established individual effects of FYM, Zn, and Mo on legume productivity, critical gaps remain in the soybean-chickpea system of rainfed Vertisols. First, only a few studies have evaluated both the direct effect on soybean and the residual effect on the succeeding chickpea through integrated nutrient management [18,20,42]. Second, comparative data on different application methods (soil, seed treatment, foliar spray) and their respective residual carryover to chickpea are lacking. Third, the system-level economic viability and protein output of these strategies have not been quantified for practical recommendations in this agro-ecology. Although individual nutrient effects are well documented, no study has simultaneously compared application methods across the directly-fertilised and residual crops. Equally, the translation of these findings into system protein productivity and economics has not been attempted. This gap is particularly important for Madhya Pradesh, where 66.9% of soils are Zn deficient [22], soil Mo is below the critical limit [36,43], and soybean-chickpea is the dominant rainfed rotation. The present study addresses these gaps through a three-year field evaluation of integrated Zn, Mo, and FYM management with comparison of application methods in Vertisols of Central India.

We hypothesised that (i) Mo would enhance soybean yield primarily through improved nodulation and biological nitrogen fixation, while Zn would act through reproductive partitioning; (ii) FYM would amplify the yield response to both Zn and Mo by improving their availability in the alkaline (pH 8.2) Vertisol; and (iii) the application method optimal for the directly-fertilised soybean would differ from the method providing the greatest residual benefit to chickpea. The objectives were to (a) assess the direct effects of integrated Zn, Mo, and FYM management on growth, nodulation, yield, and protein content of soybean; (b) quantify the residual effects on succeeding chickpea; (c) identify the optimal application method for improving system productivity, economics, and protein output; and (d) elucidate the pathways through which growth and yield characters mediate yield response using structural equation modelling.

2. Materials and methods

2.1. Experimental site and soil characteristics

Three consecutive years of field experiment (2016-17 to 2018-19) were conducted at the Research Farm of the ICAR-Indian Institute of Pulses Research, Regional Station, Bhopal, Madhya Pradesh, India (23°22′N, 77°19′E; 519 m above MSL). The site of the experiment has semi-arid conditions, which are classified under the Central Plateau and Hills agro-climatic zone of India. The zone receives an average of 1076 mm of rainfall, of which 92 percent occurs during the monsoon period [44]. The all-weather data averaged over three consecutive years during the cropping season showed the mean minimum and maximum temperatures (Fig. 1) of 6.2°C and 43°C, respectively. The mean annual rainfall during this period was 1330 mm (Fig. 3), and the peak occurs in July, coinciding with the monsoon. The relative humidity (Fig. 2) during the crop growth period varied from a minimum (44%) to a maximum (76%), indicating the semi-arid climatic conditions of the region.

Preliminary soil analysis confirmed that the soil was alkaline (pH 8.21; [45]) with a low soil organic carbon (SOC) content of 0.26% [46]. An assessment of the macronutrient status indicated that the soil was low in available nitrogen (N) (243.05 kg ha^{-1}; [47]), medium in available phosphorus (P) (14.26 kg ha^{-1}; [48]), and high in available potassium (K) (248.61 kg ha^{-1}; [49]). DTPA-extractable zinc [50] was 0.46 mg kg^{-1}, below the critical limit of 0.60 mg kg^{-1} for Indian soils [23], placing it in the deficient category. Acid ammonium acetate-EDTA extractable molybdenum was 0.016 mg kg^{-1}, well below the critical limit of 0.20 mg kg^{-1} [36,43], confirming Mo deficiency. These deficiency levels justified the application of both micronutrients in the present study.

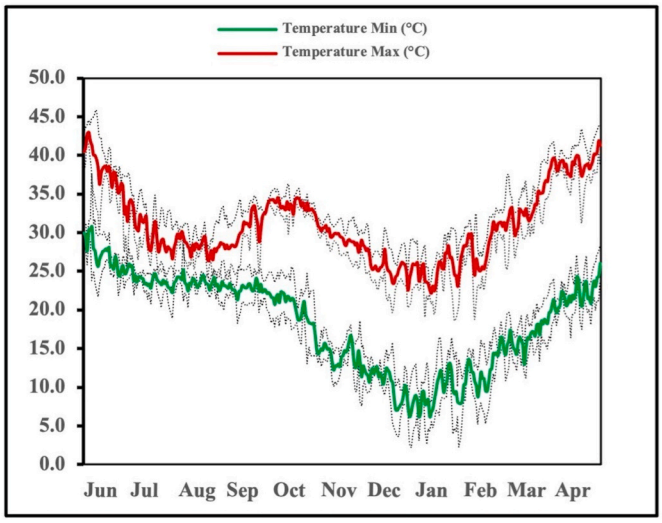


Fig. 1. Daily maximum and minimum temperatures (°C) during the cropping period (June to April) at the experimental site. Solid lines represent three-year mean values (2016-17 to 2018-19), while dotted lines indicate individual year variations.

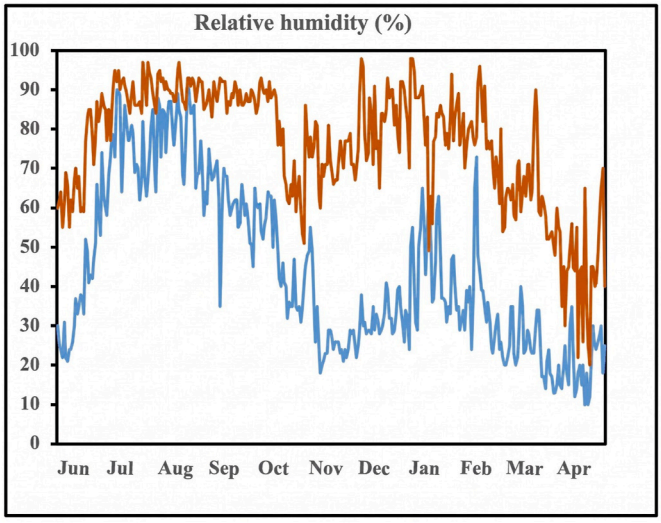


Fig. 2. Daily maximum (orange line) and minimum (blue line) relative humidity (%) during the cropping period (June to April) at the experimental site (mean of 2016-17 to 2018-19).

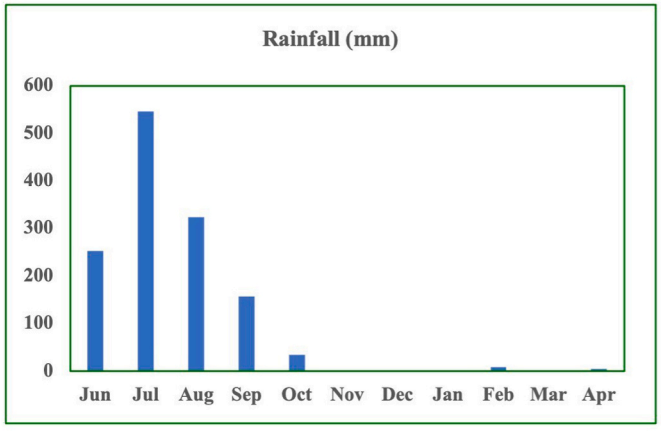


Fig. 3. Monthly rainfall (mm) during the cropping period (June to April) at the experimental site (mean of 2016-17 to 2018-19).

2.2. Treatment details and experimental layout

The experiment was designed in a split-plot design, constituting 12 treatment combinations (2 main plots x 6 subplots) with three replicates, giving a total of 36 experimental plots. Each main plot size was 8 m × 36 m, divided into six subplots of 8 m × 6 m. The main plot factor had two levels of farmyard manure (FYM), namely F0 (no FYM) and F1 (FYM at 5 t ha^{-1}). The subplot factor comprised six treatments of micronutrient management (N0 to N5): N0 (control, no micronutrient application); N1 (zinc sulphate @ 25 kg ha^{-1} as soil application); N2 (zinc sulphate @ 12.5 kg ha^{-1} as soil application + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed); N3 (ammonium molybdate @ 1 g kg^{-1} seed as seed treatment + zinc sulphate @ 25 kg ha^{-1} as soil application); N4 (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both as soil application); and N5 (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% as foliar spray). The detailed chemical specifications, elemental rates, and method and time of application are given in Table 1.

The micronutrient rates followed recommendations for soybean-based cropping systems in Vertisols of Central India. The zinc sulphate soil application rate of 25 kg ha^{-1}, foliar spray of 0.5% $ZnSO_4$, and soil application of ammonium molybdate at 0.5 kg ha^{-1} are recommended practices for soybean in this region [24]. The higher rate of 1 kg ha^{-1} in N4 follows the Mo application rates documented for chickpea in Mo-deficient soils [37,51]. The seed treatment rate of 1 g kg^{-1} seed is consistent with legume Mo seed treatment rates reported in these studies [37].

All FYM and micronutrient treatments were only imposed on the soybean crop in the kharif (rainy) season. The next crop of chickpea grown during the rabi (winter) season received only a uniform basal dose of fertilizers (20 kg N, 60 kg P_2O_5, and 40 kg K_2O ha^{-1}) to study the residual effects of the treatments applied to the soybean. For soybean, a basal application of 30 kg N, 60 kg P_2O_5, and 40 kg K_2O ha^{-1} was uniformly applied to all the plots at the time of sowing using urea (46% N), di-ammonium phosphate (46% P_2O_5), and muriate of potash (60% K_2O), respectively.

2.3. Crop management

The experimental field was prepared by one deep (25-30 cm) ploughing followed by two harrowing and planking to obtain a fine tilth and uniform seedbed. Prior to sowing, seeds of both crops were treated with carbendazim 50% WP @ 2 g kg^{-1} seed for disease management. Soybean (cv. JS-2029) was sown on 15 June 2016, 20 June 2017 and 18 June 2018 at a rate of 80 kg ha^{-1} and 30 cm row spacing. FYM and soil-applied micronutrients (Zn and Mo) were applied to soybean according to the treatment details mentioned in Section 2.2. Chickpea (cv. JG-130) was sown on 18 October 2016, 20 October 2017 and 19 October 2018 at 75 kg ha^{-1} with 30 cm row spacing. The final plant population of 30–35 plants m^{-2} was maintained in both crops.

The average maturity duration was 97 days for soybean and 122 days for chickpea over the three experimental years. Soybean received sufficient amount of rainfall throughout the growing season and had no supplemental irrigation during any of the three years. Chickpea was given one supplementary irrigation (60 mm) at the pre-flowering stage (65-70 DAS) via surface flooding. Weeds were managed by one hand weeding and one spot weeding at 25 and 45 DAS in soybean and 35 and 55 DAS in chickpea, respectively. Weekly field monitoring was performed for the timely management of pests. Pod borer (*Helicoverpa armigera*) was the major pest, particularly in chickpea, which was controlled by indoxacarb 14.5% SC at 0.5 ml L^{-1} (300 ml ha^{-1} in 600-liter water). For soybean, applications were done at 60-65 days after sowing, and for chickpea at 85-90 days after sowing. All pesticide applications were performed using a knapsack sprayer with a hollow cone nozzle to ensure uniform coverage.

2.4. Sampling procedures and measurements

All biometric observations on plant samples were taken following a systematic random sampling procedure in the net plot area, whereby all the rows situated on the border were eliminated and 0.5 m on each side was excluded to prevent border effects and ensure representative measurements.

2.4.1. Growth parameters

Ten random plants per plot were uprooted at physiological maturity to determine dry matter. Whole plants were weighed immediately on a digital electronic balance (precision ± 0.01 g) to determine fresh weight. Plant samples were dried in an oven at 70 ± 2°C for 72 h and dry weight was recorded. The amount of accumulated total dry matter was indicated as g $plant^{-1}$. Branches count per plant was determined on ten plants prior to uprooting. Only primary branches that emerged straight out of the main stem and had three or more nodes were counted as effective branches. Branches shorter than 5 cm or without reproductive structures (flowers or pods) were not counted. The values are presented as the mean number of branches per plant.

Nodulation was assessed at peak flowering (55-60 DAS in the case of soybean; 75-80 DAS in the case of chickpea), when maximum

Table 1
Treatment details, chemical formulations, elemental rates, and application methods.

Treatment code	Treatment Description & Dose	Input Material & Chemical Formula	Elemental Rate	Application Method & Timing
Main Plot Factor: Farmyard Manure (FYM) management				
F0	No FYM	—	—	—
F1	FYM @ 5 t ha^{-1}	Well-decomposed cattle manure (0.52% N, 0.26% P_2O_5, 0.48% K_2O, C:N 18.2:1)	—	Broadcast and incorporated 15 days before sowing
Subplot Factor: Micronutrient management				
N0	Control (no micronutrient application)	—	—	—
N1	Zinc sulphate @ 25 kg ha^{-1} (soil application)	Zinc sulphate heptahydrate ($ZnSO_4·7H_2O$, 21% Zn)	5.25 kg Zn ha^{-1}	Soil application, incorporated during final tillage at sowing
N2	Zinc sulphate @ 12.5 kg ha^{-1} (soil) + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed	$ZnSO_4·7H_2O$ (21% Zn) *Bacillus aryabhattai* (1×10^8 CFU ml^{-1})	2.63 kg Zn ha^{-1}	Soil application at sowing Seed treatment 24 h before sowing
N3	Ammonium molybdate @ 1 g kg^{-1} seed (seed treatment) + Zinc sulphate @ 25 kg ha^{-1} (soil)	Ammonium molybdate [$(NH_4)_6Mo_7O_{24}·4H_2O$, 52% Mo] $ZnSO_4·7H_2O$ (21% Zn)	~40 g Mo ha^{-1} 5.25 kg Zn ha^{-1}	Seed treatment 24 h before sowing Soil application at sowing
N4	Ammonium molybdate @ 1 kg ha^{-1} (soil) + Zinc sulphate @ 25 kg ha^{-1} (soil)	$(NH_4)_6Mo_7O_{24}·4H_2O$ (52% Mo) $ZnSO_4·7H_2O$ (21% Zn)	0.52 kg Mo ha^{-1} 5.25 kg Zn ha^{-1}	Soil application at sowing Soil application at sowing
N5	Ammonium molybdate @ 1 g kg^{-1} seed (seed treatment) + Zinc sulphate @ 0.5% (foliar spray)	$(NH_4)_6Mo_7O_{24}·4H_2O$ (52% Mo) $ZnSO_4·7H_2O$ (21% Zn)	~40 g Mo ha^{-1} 1.05 kg Zn ha^{-1}	Seed treatment 24 h before sowing Foliar spray (2.5 kg in 500 L water ha^{-1}) at 30 and 60 DAS

Note: Elemental Zn rates calculated from $ZnSO_4·7H_2O$ at 21% Zn; elemental Mo rates calculated from ammonium molybdate at 52% Mo. Mo seed treatment rate assumes 80 kg ha^{-1} seed rate. N5 foliar Zn = 2 sprays × 0.53 kg Zn per spray = 1.05 kg Zn ha^{-1} total.

nodulation was expected. Five plants were dug in each plot with the utmost care, with a spade in a radius of 20 cm around the base of the plant to a depth of 20 cm to obtain the maximum number of nodule-bearing roots. The root systems were carefully rinsed with a 2 mm mesh sieve filter to remove the adhering soil without detaching the nodules. The number of nodules was counted and grouped into effective (pink to red internal colour to show active nitrogen fixation) and ineffective (white to green internal colour), and carefully removed from the roots and transferred to pre-weighed (analytical balance with precision 0.001 g) aluminum foil packets, where they were dried in an oven at 70 ± 2°C for 72 h. Results are reported as mean nodule number per plant and nodule dry weight (mg $plant^{-1}$). This manual excavation and sieve-washing procedure is a standard field technique for legume nodule recovery [52]. The same protocol was applied uniformly across all plots and treatments to ensure comparability.

2.4.2. Yield attributes

The same ten plants used for dry matter determination were used to record yield attributes at physiological maturity. Pod count per plant was performed on standing plants immediately before uprooting. The number of seeds per plant was measured by threshing of individual plant samples and counting the number of seeds. For chickpea, seeds per pod were additionally calculated by dividing the total seeds per plant by the number of pods per plant. The weight of 100 seeds was calculated using eight random samples of 100 seeds each picked out of the bulk seed lot of each plot following cleaning and drying to a moisture content of about 12%. An electronic balance (precision ± 0.01 g) was used to weigh the seeds and computed the mean weight of 100 seeds. In cases where the moisture content in the seed was higher than 12%, the weights were adjusted according to the standard moisture correction formula.

2.4.3. Grain yield and harvest

At physiological maturity, all net plot areas (excluding border rows and end margins) were manually harvested at ground level. Plants that were harvested were tied in bundles and dried in the field over 5-7 days. The bundles were then threshed manually on a clean threshing floor with wooden sticks, and the seeds were separated by winnowing. A platform balance was used to weigh the clean seeds, and the grain yield was adjusted to 12% moisture content. The final grain yield was measured in kg ha^{-1}.

2.4.4. System productivity

The system productivity was calculated on a soybean equivalent basis using the price equivalence method to assess the overall productivity of the soybean–chickpea cropping system:

$$SP = Ys + (Yc \times Pc / Ps) \quad (1)$$

where: *SP* = System productivity expressed as soybean equivalent yield (kg ha^{-1}) *Ys* = Soybean grain yield (kg ha^{-1}) *Yc* = Chickpea grain yield (kg ha^{-1}) *Pc* = Price of chickpea (₹ kg^{-1}) *Ps* = Price of soybean (₹ kg^{-1}).

2.5. Economic analysis

Economic parameters were calculated to assess the profitability of the system. Net returns were calculated as follows:

$$\text{Net returns (₹ ha}^{-1}\text{)} = \text{Gross returns} - \text{Production cost.} \quad (2)$$

Gross returns were calculated based on the minimum support price (MSP) declared by the Government of India for the respective crop years. Production costs were computed treatment-wise, including cost of all inputs, labour, transportation and machinery at prevailing local rates.

The benefit-cost ratio was determined as follows:

$$\text{B:C ratio} = \text{Net returns (₹ ha}^{-1}\text{)} / \text{Production cost (₹ ha}^{-1}\text{)} \quad (3)$$

Daily profitability was calculated as follows:

$$\text{Profitability (₹ ha}^{-1}\text{ day}^{-1}\text{)} = \text{Net returns (₹ ha}^{-1}\text{)} / \text{Total crop duration (days)} \quad (4)$$

The total crop duration was 219 days, comprising soybean (97 days) and chickpea (122 days), based on the mean maturity period across three years.

2.6. Protein productivity and nutritional assessment

Seed nitrogen content was determined using the Kjeldahl method [53], and protein content was calculated by multiplying the nitrogen percentage with a conversion factor of 6.25 for both crops.

The protein productivity for each crop was calculated as follows:

$$\text{Protein productivity (kg ha}^{-1}\text{)} = [\text{grain yield (kg ha}^{-1}\text{)} \times \text{protein content (\%)}] / 100 \quad (5)$$

The system protein productivity was calculated as follows:

$$\text{System protein productivity (kg ha}^{-1}\text{)} = \text{Soybean protein productivity} + \text{Chickpea protein productivity} \quad (6)$$

The protein production efficiency was calculated as follows:

$$\text{Protein production efficiency (kg ha}^{-1}\text{ day}^{-1}\text{)} = \text{System protein productivity / Crop duration} \quad (7)$$

The protein supply capacity (persons ha^{-1} $year^{-1}$), representing the number of persons whose protein requirement could theoretically be met by the protein produced per hectare based on the recommended dietary allowance of 58 g $person^{-1}$ day^{-1} [54], was calculated as:

$$\text{Persons ha}^{-1}\text{ year}^{-1} = [\text{System protein productivity (kg ha}^{-1}\text{)} \times 1000] / [58\text{ (g person}^{-1}\text{ day}^{-1}\text{)} \times 365\text{ (days year}^{-1}\text{)}] \quad (8)$$

This calculation assumes all grain protein is available for human consumption. It does not account for protein quality (amino acid composition or digestibility), post-harvest and processing losses, or end-use allocation (oil extraction, animal feed).

2.7. Data analysis

Data from the three years were subjected to a pooled split-plot analysis of variance (ANOVA) with year as a random factor, FYM as the fixed main-plot factor, and micronutrient management as the fixed sub-plot factor, with three replications within each year. The ANOVA was computed with two error terms: Error (a), derived from the year x replication × FYM interaction, for testing the main-plot effect, and Error (b), derived from the residual, for testing the sub-plot and interaction effects. Year × FYM, year × micronutrient, and year × FYM × micronutrient interactions were tested. For seed yield in both crops, all year × treatment interactions were non-significant ($p > 0.05$). For some growth parameters (seeds per plant, test weight, nodules per plant) and protein productivity, significant Year × FYM interactions were detected ($p < 0.05$; Tables 2, 3, and 5); however, in all cases the direction and ranking of treatment effects remained consistent across years, with only the magnitude of response varying between seasons. This consistency justified the presentation of pooled means across the three years. Treatment means were compared using Fisher's LSD test ($p < 0.05$); means followed by the same letter within a column are not significantly different. Two-way hierarchical cluster analysis was performed using Ward's method with Euclidean distance in Python 3 (SciPy 1.16.3, scikit-learn 1.7.2, Matplotlib 3.10.7). Quadratic regression was used to model the relationship between nutrient management and protein content, with R^2 determined separately for each FYM level. Structural equation modelling (SEM) was performed using the lavaan package [55] in R (version 4.3.0).

3. Results

3.1. Growth, yield attributes and yield of soybean

Application of FYM at 5t ha^{-1} (F1) significantly improved all measured parameters compared to the control (F0) ($p < 0.05$). The F1 treatment had 20.9% greater dry matter accumulation, 23.2% more branches per plant, 21.2% higher nodules per plant, and 27.7% greater nodule dry weight (Table 2). The reproductive parameters also improved, with F1 showing 23.6% more pods per plant and 16.2% more seeds per plant and 6.2% higher test weight ($p < 0.05$). As a result, the seed yield in F1 (1403 kg ha^{-1}) was 25.8% higher than that in F0 ($p < 0.05$, Table 2).

Micronutrient management strategies showed a significant improvement in soybean performance (Table 2). The combined

Table 2
Growth, yield attributes and yield of soybean crop under farm yard manure and micro-nutrient management in Vertisol of India (pooled of three years).

Treatments	Dry matter at harvest (g $plant^{-1}$)	Branches $plant^{-1}$	Nodules $plant^{-1}$	Nodule dry weight (mg $plant^{-1}$)	Pods $plant^{-1}$	Seeds $plant^{-1}$	Test weight (g)	Seed yield (kg ha^{-1})
FYM management								
F0	18.95 ± 0.30^{b}	3.02 ± 0.11^{b}	32.68 ± 0.68^{b}	279.31 ± 8.32^{b}	27.43 ± 1.39^{b}	52.74 ± 2.51^{b}	12.62 ± 0.14^{b}	1115 ± 41.40^{b}
F1	22.92 ± 0.34^{a}	3.72 ± 0.12^{a}	39.60 ± 0.66^{a}	356.57 ± 7.29^{a}	33.90 ± 1.51^{a}	61.29 ± 2.84^{a}	13.40 ± 0.13^{a}	1403 ± 43.30^{a}
Micronutrient management								
N0	18.11 ± 0.59^{c}	2.62 ± 0.14^{c}	30.56 ± 1.28^{d}	265.28 ± 13.70^{c}	23.96 ± 2.19^{c}	45.39 ± 3.91^{d}	11.98 ± 0.20^{c}	1016 ± 73.27^{c}
N1	20.19 ± 0.54^{b}	3.08 ± 0.17^{b}	34.22 ± 1.17^{c}	299.14 ± 13.31^{b}	28.26 ± 2.34^{b}	52.74 ± 4.22^{c}	12.67 ± 0.20^{b}	1197 ± 75.63^{b}
N2	20.48 ± 0.73^{b}	3.20 ± 0.17^{b}	34.67 ± 1.21^{c}	307.98 ± 13.97^{b}	29.17 ± 2.39^{b}	54.76 ± 4.54^{c}	12.70 ± 0.22^{b}	1213 ± 74.85^{b}
N3	21.93 ± 0.59^{a}	3.72 ± 0.21^{a}	38.35 ± 1.14^{b}	340.77 ± 14.74^{a}	33.42 ± 2.61^{a}	61.16 ± 4.70^{b}	13.45 ± 0.21^{a}	1357 ± 76.56^{a}
N4	22.30 ± 0.65^{a}	3.73 ± 0.22^{a}	39.77 ± 1.27^{a}	350.47 ± 15.88^{a}	34.19 ± 2.73^{a}	63.27 ± 4.87^{ab}	13.64 ± 0.21^{a}	1377 ± 77.00^{a}
N5	22.59 ± 0.74^{a}	3.86 ± 0.23^{a}	39.29 ± 1.16a^{b}	344.00 ± 17.85^{a}	34.99 ± 2.72^{a}	64.75 ± 5.03^{a}	13.64 ± 0.23^{a}	1393 ± 78.43^{a}
Year								
2016-17	20.95 ± 0.37a^{b}	2.96 ± 0.10^{b}	33.90 ± 0.62^{b}	271.43 ± 10.14^{c}	20.19 ± 0.52^{c}	38.63 ± 0.84^{c}	13.83 ± 0.14^{a}	957 ± 36.29^{c}
2017-18	20.38 ± 0.73^{b}	2.88 ± 0.08^{b}	34.65 ± 1.26^{b}	307.81 ± 7.29^{b}	28.96 ± 1.22^{b}	51.85 ± 1.95^{b}	12.38 ± 0.15^{c}	1228 ± 31.38^{b}
2018-19	21.47 ± 0.35^{a}	4.27 ± 0.14^{a}	39.88 ± 0.69^{a}	374.57 ± 9.66^{a}	42.85 ± 1.11^{a}	80.55 ± 1.86^{a}	12.83 ± 0.15^{b}	1591 ± 43.07^{a}
ANOVA								
FYM x Micronutrient management	0.4685	0.0997	0.9813	0.2075	0.4656	0.6284	0.9987	0.9435
Year	0.0124	0.0285	<0.001	0.0154	0.008	<0.001	<0.001	0.0174
Year x FYM	0.0831	0.9508	0.1432	0.1014	0.2376	0.0004	0.0376	0.5243
Year x Micronutrient management	0.0793	0.0571	0.0421	0.1842	0.0602	0.0410	0.7413	0.9056
Year x FYM x Micronutrient management	0.0914	0.6283	0.2205	0.5363	0.0773	0.0667	0.9999	0.6485

F0 (no FYM); F1 (FYM at 5t ha^{-1}); N0 (control, no micronutrient application); N1 (zinc sulphate @ 25 kg ha^{-1} as soil application); N2 (zinc sulphate @ 12.5 kg ha^{-1} as soil application + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed); N3 (ammonium molybdate @ 1 g kg^{-1} seed as seed treatment + zinc sulphate @ 25 kg ha^{-1} as soil application); N4 (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both as soil application); and N5 (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% as foliar spray); within column mean (mean ± standard error of mean) values with different lowercase letter (a-d) indicate significant difference at $p < 0.05$). Values in the ANOVA section are p-values. '< 0.001' indicates significance at $p < 0.001$.

Table 3
Growth, yield attributes and yield of chickpea crop under residual effect of farm yard manure and micro-nutrient management in Vertisol of India (pooled of three years).

Treatments	Dry matter at harvest (g plant^{-1})	Branches plant^{-1}	Nodules plant^{-1}	Nodule dry weight (mg plant^{-1})	Pods plant^{-1}	Seeds plant^{-1}	Test weight (g)	Seed yield (kg ha^{-1})
FYM management								
F0	23.32 ± 0.76^{b}	3.31 ± 0.08^{b}	30.09 ± 0.87^{b}	200.46 ± 9.77^{b}	45.71 ± 1.62^{b}	63.48 ± 0.97^{b}	22.59 ± 0.16^{b}	1766 ± 31.35^{b}
F1	28.32 ± 0.84^{a}	4.06 ± 0.09^{a}	36.08 ± 0.79^{a}	291.63 ± 10.64^{a}	52.76 ± 1.81^{a}	76.84 ± 1.44^{a}	25.35 ± 0.27^{a}	2105 ± 38.47^{a}
Micronutrient management								
N0	20.61 ± 1.35^{c}	3.05 ± 0.14^{c}	28.13 ± 1.73^{c}	167.60 ± 13.22^{c}	41.49 ± 2.48^{c}	59.01 ± 1.89^{c}	22.36 ± 0.38^{d}	1630 ± 56.99^{d}
N1	23.97 ± 1.33^{b}	3.50 ± 0.15^{b}	31.69 ± 1.50^{b}	226.42 ± 14.51^{b}	46.86 ± 2.84^{b}	67.26 ± 2.15^{b}	23.53 ± 0.44^{c}	1851 ± 74.17^{c}
N2	24.78 ± 1.51^{b}	3.51 ± 0.14^{b}	31.72 ± 1.61^{b}	228.75 ± 14.21^{b}	46.03 ± 2.58^{b}	66.84 ± 2.11^{b}	23.56 ± 0.47^{c}	1861 ± 62.49^{c}
N3	28.58 ± 1.37^{a}	4.00 ± 0.15^{a}	35.41 ± 1.19^{a}	283.20 ± 25.87^{a}	52.58 ± 3.06^{a}	75.33 ± 2.05^{a}	24.94 ± 0.45^{ab}	2106 ± 57.03^{ab}
N4	29.09 ± 1.41^{a}	4.04 ± 0.14^{a}	36.06 ± 1.35^{a}	287.02 ± 20.35^{a}	54.16 ± 3.10^{a}	77.48 ± 2.59^{a}	25.09 ± 0.52^{a}	2138 ± 55.98^{a}
N5	27.92 ± 1.10^{a}	4.02 ± 0.15^{a}	35.50 ± 1.45^{a}	283.30 ± 18.72^{a}	54.27 ± 3.41^{a}	75.04 ± 2.38^{a}	24.35 ± 0.50^{b}	2027 ± 57.40^{b}
Year								
2016-17	28.40 ± 0.73^{b}	3.23 ± 0.08^{c}	38.00 ± 0.64^{a}	271.22 ± 20.89^{a}	33.78 ± 0.78^{c}	67.68 ± 1.61^{b}	25.22 ± 0.42^{a}	1798 ± 45.77^{c}
2017-18	19.51 ± 0.86^{c}	3.80 ± 0.13^{b}	35.09 ± 0.68^{b}	201.35 ± 9.06^{b}	59.11 ± 1.44^{a}	67.30 ± 1.63^{b}	23.07 ± 0.24^{c}	2059 ± 58.03^{a}
2018-19	29.56 ± 0.67^{a}	4.03 ± 0.10^{a}	26.16 ± 0.92^{c}	265.57 ± 6.88^{a}	54.81 ± 1.13^{b}	75.50 ± 2.04^{a}	23.62 ± 0.29^{b}	1950 ± 40.17^{b}
ANOVA								
FYM x Micronutrient management	0.5728	0.8946	0.7033	0.2641	0.5808	0.3718	0.5815	0.8583
Year	<0.001	0.0052	0.0245	<0.001	0.0352	0.0412	0.0061	0.0087
Year x FYM	0.7896	0.0157	0.0669	0.0941	0.1086	0.2106	0.2561	0.3768
Year x Micronutrient management	0.3799	0.8458	0.0457	0.0787	0.0845	0.9435	0.7227	0.7929
Year x FYM x Micronutrient management	0.3365	0.5165	0.3566	0.947	0.2067	0.7053	0.9587	0.8591

F0 (no FYM); F1 (FYM at 5t ha^{-1}); N0 (control, no micronutrient application); N1 (zinc sulphate @ 25 kg ha^{-1} as soil application); N2 (zinc sulphate @ 12.5 kg ha^{-1} as soil application + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed); N3 (ammonium molybdate @ 1 g kg^{-1} seed as seed treatment + zinc sulphate @ 25 kg ha^{-1} as soil application); N4 (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both as soil application); and N5 (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% as foliar spray); within column mean (mean ± standard error of mean) values with different lowercase letter (a-d) indicate significant difference at $p < 0.05$). Values in the ANOVA section are p-values. '< 0.001' indicates significance at $p < 0.001$.

application of Mo and Zn (N3, N4, and N5) consistently improved all growth parameters, yield attributes and yield compared to the single applications of Zn (N1 and N2) and control (N0) ($p < 0.05$). The N5 treatment (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% foliar spray) produced the greatest seed yield of 1393 kg ha^{-1}, which was at par with N4 and N3, corresponding to yield increases of 37.1%, 35.6%, and 33.6%, respectively, compared to N0 ($p < 0.05$, Table 2). In the single zinc application treatments (N1 and N2), intermediate performance was recorded in terms of seed yield, representing 17.8% and 19.4% improvements over the control (N0).

3.2. Residual effect on growth, yield attributes and yield of chickpea

The residual effect of FYM application @ 5 t ha^{-1} (F1) in the previous soybean crop continued to significantly improve all the measured parameters in chickpea as compared to the control (F0) ($p < 0.05$). The F1 treatment registered 21.4% superior accumulation of dry matter, 22.7% greater branches per plant, 19.9% more nodules per plant, and 45.5% higher dry weight of the nodules (Table 3). The reproductive parameters also showed considerable improvements, with F1 giving 15.4% more pods, 21.0% more seeds per plant, and 12.2% higher test weight ($p < 0.05$). Consequently, the seed yield in F1 (2105 kg ha^{-1}) was 19.2% higher than that in F0 ($p < 0.05$, Table 3), indicating the sustained benefits of organic matter addition.

The residual impact of the micronutrient management strategies used on soybean showed distinct performances in chickpea (Table 3). Combined residual application of Mo and Zn (N3, N4, and N5) was superior to single Zn applications (N1 and N2) ($p < 0.05$). The N4 treatment had the highest seed yield of 2138 kg ha^{-1} (31.2% over N0), followed by N3 (29.2%) and N5 (24.3%) ($p < 0.05$, Table 3). Single zinc application treatments (N1 and N2) had medium residual effects, which were 13.6 and 14.2% higher than the control, respectively, indicating that dual micronutrient application treatments have a more persistent effect in cropping systems.

3.3. System productivity and economic profitability

Application of FYM at 5 t ha^{-1} (F1) significantly improved system productivity and economic return ($p < 0.05$) compared to no FYM application (F0). The system productivity (soybean equivalent yield) in F1 (4439 kg ha^{-1}) was 21.2% higher than that in F0 (Table 4). This increased productivity was translated into improved economic parameters. The net return in F1 (₹ 94,052 ha^{-1}) was 33.4% higher than F0 ($p < 0.05$). The B:C ratio improved by 31.1% in F1 compared to that in F0 ($p < 0.05$). Similarly, daily profitability in F1 (₹ 429.46 ha^{-1} day^{-1}) was 33.4% higher in comparison to F0 ($p < 0.05$).

Micronutrient management strategies showed progressive increases in system productivity and economic profitability (Table 4). The combined application of Mo and Zn (N3, N4, and N5) consistently performed better than the single applications of Zn (N1 and N2) for all economic parameters ($p < 0.05$). The N4 treatment (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both in soil application) resulted in the highest system productivity (4464 kg ha^{-1}) and net return (₹94,332 ha^{-1}), which was 32.7 and 54.7% higher than the yield and net return in N0 ($p < 0.05$, Table 4). The N4 treatment also showed the highest B:C ratio (3.52) and daily profitability (₹ 430.74 ha^{-1} day^{-1}) which showed an improvement of 53.0% and 54.7%, respectively, compared to N0 ($p < 0.05$). Single zinc application treatments (N1 and N2) had an intermediate economic performance. The N2 treatment (zinc sulphate @ 12.5 kg ha^{-1} + Zn-solubilizing bacteria) recorded system productivity of 3897 kg ha^{-1}, net return of ₹77,417 ha^{-1}, and B:C ratio of 2.91, representing increases of 15.9%, 27.0%, and 26.5% over N0, respectively ($p < 0.05$).

Table 4
System productivity (soybean equivalent yield) and economic profitability of soybean-chickpea cropping system as influenced by FYM and micronutrient management in Vertisol of India (pooled of three years).

Treatments	System productivity (kg ha^{-1})	Net return (₹ ha^{-1})	B:C ratio	Profitability (₹ ha^{-1} day^{-1})
FYM management				
F0	3662 ± 66.95^{b}	70512 ± 2776^{b}	2.67 ± 0.08^{b}	321.98 ± 12.68^{b}
F1	4439 ± 73.19^{a}	94052 ± 3161^{a}	3.50 ± 0.08^{a}	429.46 ± 14.44^{a}
Micronutrient management				
N0	3364 ± 122.98^{c}	60979 ± 4965^{c}	2.30 ± 0.14^{c}	278.44 ± 22.67^{c}
N1	3864 ± 145.75^{b}	75702 ± 5831^{b}	2.83 ± 0.16^{b}	345.67 ± 26.63^{b}
N2	3897 ± 127.43^{b}	77417 ± 5126^{b}	2.91 ± 0.14^{b}	353.50 ± 23.41^{b}
N3	4396 ± 113.90^{a}	93068 ± 5117^{a}	3.48 ± 0.13^{a}	424.97 ± 23.37^{a}
N4	4464 ± 108.90^{a}	94332 ± 4946^{a}	3.52 ± 0.12^{a}	430.74 ± 22.59^{a}
N5	4320 ± 123.61^{a}	92195 ± 5204^{a}	3.48 ± 0.14^{a}	420.98 ± 23.76^{a}
Year				
2016-17	3711 ± 100.53^{b}	61368 ± 2787^{c}	2.72 ± 0.12^{c}	280.22 ± 12.73^{c}
2017-18	4199 ± 110.07^{a}	83992 ± 3373^{b}	3.09 ± 0.12^{b}	383.53 ± 15.41^{b}
2018-19	4242 ± 89.96^{a}	101487 ± 3133^{a}	3.44 ± 0.10^{a}	463.41 ± 14.31^{a}
ANOVA				
FYM x Micronutrient management	0.7971	0.8104	0.8413	0.8104
Year	0.0008	0.0074	<0.001	0.0047
Year x FYM	0.2256	0.0636	0.0978	0.1423
Year x Micronutrient management	0.6609	0.9254	0.4227	0.9254
Year x FYM x Micro-nutrient management	0.977	0.9604	0.9712	0.9604

F0 (no FYM); F1 (FYM at 5t ha^{-1}); N0 (control, no micronutrient application); N1 (zinc sulphate @ 25 kg ha^{-1} as soil application); N2 (zinc sulphate @ 12.5 kg ha^{-1} as soil application + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed); N3 (ammonium molybdate @ 1 g kg^{-1} seed as seed treatment + zinc sulphate @ 25 kg ha^{-1} as soil application); N4 (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both as soil application); and N5 (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% as foliar spray); within column mean (mean ± standard error of mean) values with different lowercase letter (a-d) indicate significant difference at $p < 0.05$). Values in the ANOVA section are p-values. '< 0.001' indicates significance at $p < 0.001$.

Table 5
Protein productivity and protein supply capacity of soybean-chickpea cropping system as influenced by FYM and micronutrient management in Vertisol of India (pooled of three years).

Treatments	Soybean protein productivity (kg ha^{-1})	Chickpea protein productivity (kg ha^{-1})	System protein productivity (kg ha^{-1})	Protein production efficiency (kg ha^{-1} day^{-1})	Protein supply capacity (Persons ha^{-1} year^{-1})
FYM management					
F0	438.67 ± 17.10^{b}	380.82 ± 9.07^{b}	819.49 ± 23.34^{b}	3.74 ± 0.11^{b}	38.71 ± 1.10^{b}
F1	592.22 ± 19.16^{a}	501.92 ± 10.64^{a}	1094.14 ± 26.00^{a}	5.00 ± 0.12^{a}	51.68 ± 1.23^{a}
Micronutrient management					
N0	393.36 ± 28.76^{c}	342.47 ± 15.64^{d}	735.83 ± 40.91^{c}	3.36± 0.19^{c}	34.76 ± 1.93^{c}
N1	482.84 ± 33.92^{b}	412.28 ± 21.87^{c}	895.13 ± 51.55^{b}	4.09 ± 0.24^{b}	42.28 ± 2.44^{b}
N2	490.92 ± 33.01^{b}	414.59 ± 19.04^{c}	905.51 ± 46.96^{b}	4.13 ± 0.21^{b}	42.77 ± 2.22^{b}
N3	567.04 ± 33.45^{a}	494.54 ± 15.01a^{b}	1061.58 ± 41.91^{a}	4.85 ± 0.19^{a}	50.15 ± 1.98^{a}
N4	574.34 ± 35.74^{a}	506.74 ± 17.64^{a}	1081.08 ± 46.40^{a}	4.94 ± 0.21^{a}	51.07 ± 2.19^{a}
N5	584.16 ± 34.26^{a}	477.60 ± 18.39^{b}	1061.76 ± 46.64^{a}	4.85 ± 0.21^{a}	50.15 ± 2.20^{a}
Year					
2016-17	392.95 ± 17.61^{c}	410.18 ± 14.40^{c}	803.13 ± 30.67^{c}	3.67 ± 0.14^{c}	37.94 ± 1.45^{c}
2017-18	502.47 ± 16.10^{b}	469.71 ± 17.58^{a}	972.18 ± 32.58^{b}	4.44 ± 0.15^{b}	45.92 ± 1.54^{b}
2018-19	650.92 ± 21.83^{a}	444.22 ± 13.82^{b}	1095.14 ± 34.04^{a}	5.00 ± 0.16^{a}	51.73 ± 1.61^{a}
ANOVA					
FYM x Micronutrient management	0.4659	0.2436	0.0429	0.0429	0.0429
Year	0.0043	<0.001	<0.001	<0.001	<0.001
Year x FYM	0.1735	0.0033	0.0309	0.0309	0.0309
Year x Micronutrient management	0.7728	0.8607	0.9221	0.9221	0.9221
Year x FYM x Micronutrient management	0.4982	0.8493	0.8735	0.8735	0.8735

F0 (no FYM); F1 (FYM at 5t ha^{-1}); N0 (control, no micronutrient application); N1 (zinc sulphate @ 25 kg ha^{-1} as soil application); N2 (zinc sulphate @ 12.5 kg ha^{-1} as soil application + Zn-solubilizing bacteria @ 3 ml kg^{-1} seed); N3 (ammonium molybdate @ 1 g kg^{-1} seed as seed treatment + zinc sulphate @ 25 kg ha^{-1} as soil application); N4 (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}, both as soil application); and N5 (ammonium molybdate @ 1 g kg^{-1} seed treatment + zinc sulphate @ 0.5% as foliar spray); within column mean (mean ± standard error of mean) values with different lowercase letter (a-d) indicate significant difference at $p < 0.05$). Values in the ANOVA section are p-values. '< 0.001' indicates significance at $p < 0.001$.

3.4. Protein productivity and protein supply capacity

FYM application at 5 t ha^{-1} significantly improved the protein productivity of both crops compared to the control ($p < 0.05$). Soybean protein productivity improved by 35.0% and chickpea by 31.8% under FYM application (Table 5). As a result, the system protein productivity increased by 33.5% with FYM application, coupled with a 33.7% increase in protein production efficiency ($p < 0.05$). The FYM-applied plots showed 33.5% greater protein supply capacity (persons ha^{-1} year^{-1}) compared to plots without FYM application ($p < 0.05$, Table 5).

Micronutrient management strategies showed progressive improvements in protein productivity parameters (Table 5). The combined application of Mo and Zn consistently had better effects than the single application of Zn in all protein-related measures ($p < 0.05$). The N4

treatment (ammonium molybdate @ 1 kg ha^{-1} + zinc sulphate @ 25 kg ha^{-1}) recorded the greatest system protein productivity at 1081.08 kg ha^{-1} which was an increase of 46.9% compared to the control ($p < 0.05$). This treatment also recorded the maximum protein production efficiency of 4.94 (kg ha^{-1} day^{-1}) and thus had a protein supply capacity of 51.07 persons ha^{-1} year^{-1}, which was 46.9% higher than the control ($p < 0.05$, Table 5). The N3 and N5 treatments had similar system protein productivity with a protein supply capacity of approximately 50 persons ha^{-1} year^{-1} ($p < 0.05$). Single zinc application treatments (N1 and N2) showed intermediate protein productivity, which increased by 21.7% to 23.1% over the control ($p < 0.05$). The FYM × micronutrient interaction was significant for system protein productivity, protein production efficiency, and protein supply capacity ($p = 0.04$, Table 5), indicating that the protein enhancement from micronutrient application was greater under FYM-amended conditions.

3.5. Inter-relationship and multivariate analysis of yield and quality attributes

Correlation matrices for soybean (Fig. 4) and chickpea (Fig. 5) showed a strong positive correlation between seed yield and protein productivity ($r = 0.99$ and 0.97, respectively; $p < 0.001$), with the regression analysis (Fig. 6) further confirming that protein content did not decline with increasing yield. In the upper triangle of the correlation plots, the split-treatment analysis showed that the association between growth parameters and yield components was generally strengthened through FYM (F1) application compared to F0. Specifically, pods per plant in soybean ($r = 0.90$, $p < 0.001$) and seed-related characteristics in chickpea were the major contributors to productivity.

The use of FYM and micronutrient management improved the yield attributes and protein content of both soybean (Table S3) and chickpea (Table S4). Regression analysis indicated that soybean protein (Fig. 6a) was quadratic, with higher coefficients under FYM application ($R^2 = 0.83$) than in the control ($R^2 = 0.81$) (Table S2). Similarly, chickpea protein content (Fig. 6b) showed a strong quadratic response under both F0 ($R^2 = 0.91$) and F1 ($R^2 = 0.84$). Both crops had diminishing returns at higher levels of micronutrients under F1, as shown by the negative coefficients of the quadratic functions in the regression equations.

Hierarchical clustering and heatmap analyses for soybean (Fig. 7a) and chickpea (Fig. 7b) created categories of combinations of treatments based on their performance in all growth and yield traits. In both crops, the treatment combinations F1N3, F1N4, and F1N5 were grouped together and showed the highest intensity (dark red) for yield, B:C ratio, and protein productivity. In contrast, the F0N0, F1N0, F0N1, and F0N2 treatments formed a cluster with the lowest values (dark blue and light blue), indicating that without combined Zn + Mo application, performance remained low regardless of FYM.

3.6. Identification of decisive yield attributes through structural equation modelling

Structural equation modelling (SEM) revealed that yield characteristics (YC) were the decisive factor influencing seed yield in both soybean and chickpea across FYM treatments (Fig. 8a and b; Fig. 9a and b; Table S1). In soybean, the direct effect of growth characters (GC) on yield was non-significant under both F0 ($\beta = 0.17$) and F1 ($\beta = 0.01$), whereas YC exhibited highly significant positive effects ($\beta = 1.0$ and 1.14, respectively; $p < 0.001$). The model explained $R^2 = 0.57$ (F0) and $R^2 = 0.74$ (F1) of the variance in soybean yield, with FYM application strengthening the GC-YC covariance (0.33 to 0.43). In chickpea, GC showed non-significant negative effects on yield under both F0 ($\beta = -0.05$) and F1 ($\beta = -0.21$), whereas YC demonstrated strong positive effects ($\beta = 1.08$ and 1.06, respectively; $p < 0.001$). The GC-YC covariance was notably higher in chickpea (0.75–0.78) compared to soybean. The model explained $R^2 = 0.60$ (F0) and $R^2 = 0.34$ (F1) of the variance in chickpea yield, suggesting that FYM application introduces additional unmeasured factors affecting yield determination in chickpea.

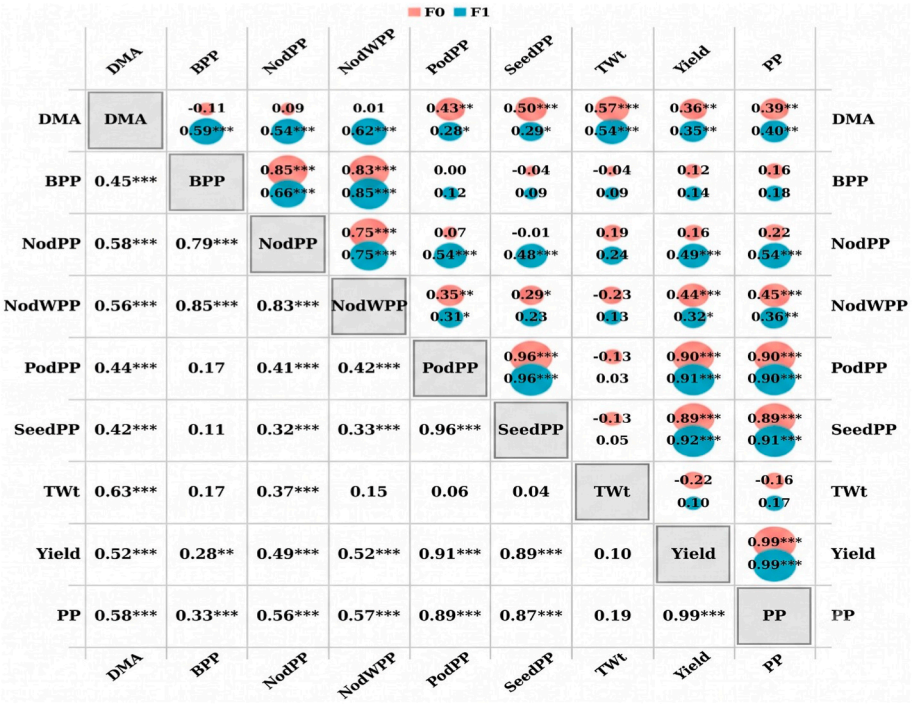


Fig. 4. Correlation matrix of growth, yield attributes, and quality parameters of soybean without the application of FYM (F0) and FYM application (F1). Lower triangle shows overall correlation coefficients, upper triangle shows split correlations for F0 (red) and F1 (blue). DMA: dry matter accumulation; BPP: branches per plant; NodPP: nodules per plant; NodWPP: nodule weight per plant; PodPP: pods per plant; SeedPP: seeds per plant; TWt: test weight; PP: protein productivity; Yield: grain yield. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

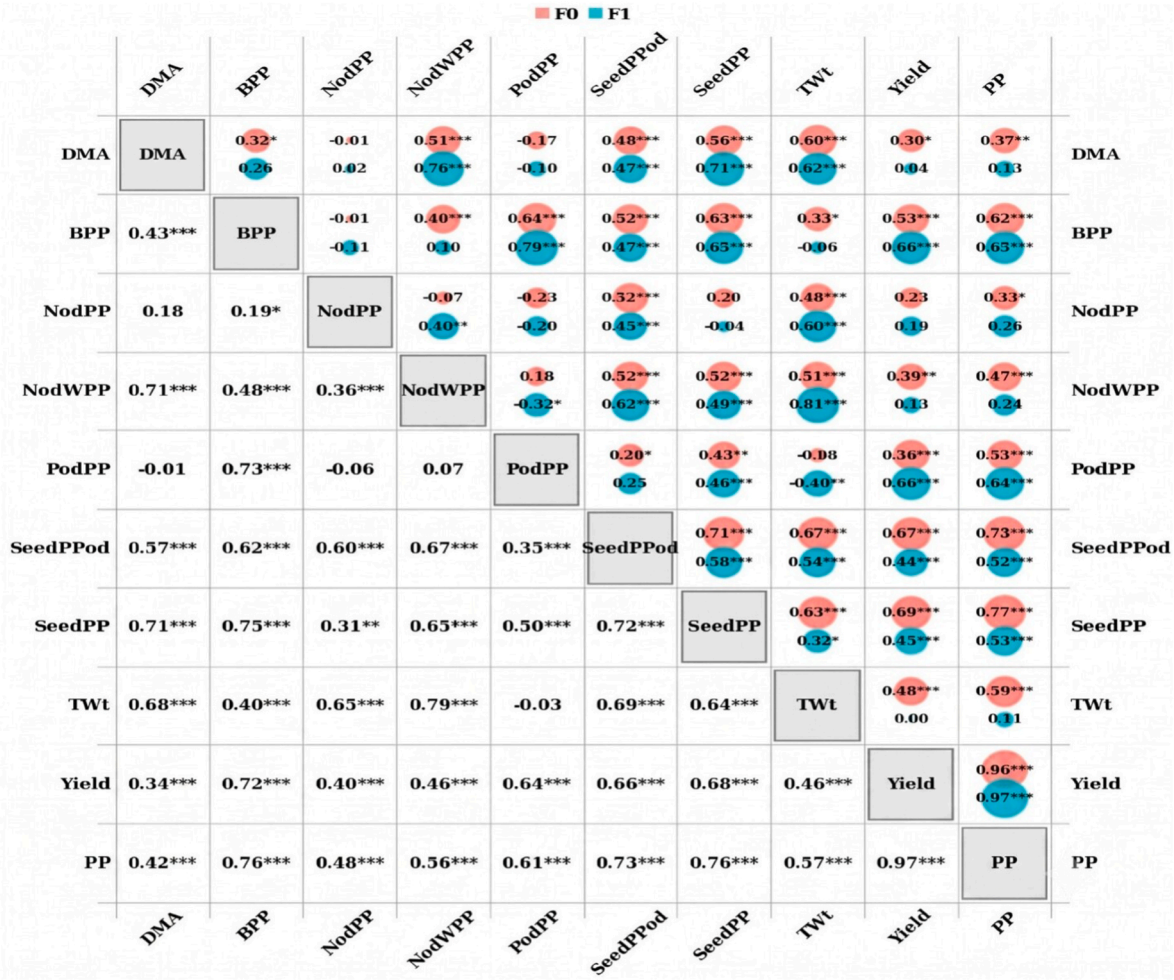


Fig. 5. Correlation matrix of growth, yield attributes, and quality parameters of chickpea without the application of FYM (F0) and FYM application (F1). Lower triangle shows overall correlation coefficients, upper triangle shows split correlations for F0 (red) and F1 (blue). DMA: dry matter accumulation; BPP: branches per plant; NodPP: nodules per plant; NodWPP: nodule weight per plant; PodPP: pods per plant; SeedPPod: seeds per pod; SeedPP: seeds per plant; TWt: test weight; PP: protein productivity; Yield: grain yield. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

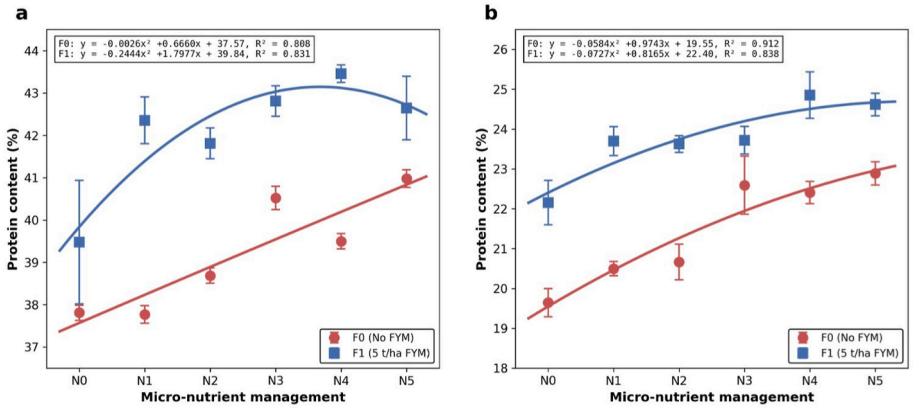


Fig. 6. Quadratic regression of protein content (%) as influenced by FYM management without the application of FYM (F0, red) and FYM application (F1, blue) in (a) soybean and (b) chickpea. N0: control; N1: $ZnSO_4$ @ 25 kg ha^{-1} soil application; N2: $ZnSO_4$ @ 12.5 kg ha^{-1} + Zn-solubilising bacteria; N3: ammonium molybdate @ 1 g kg^{-1} seed treatment + $ZnSO_4$ @ 25 kg ha^{-1} soil application; N4: ammonium molybdate @ 1 kg ha^{-1} + $ZnSO_4$ @ 25 kg ha^{-1}, both as soil application; N5: ammonium molybdate @ 1 g kg^{-1} seed treatment + $ZnSO_4$ @ 0.5% foliar spray. Error bars indicate standard error of mean.

4. Discussion

India has about 25% of the world's Vertisol area, and these soils are mostly found in semi-arid tropical areas under rainfed cultivation [56]. Vertisols have unique agronomic challenges, such as poor drainage, hard-setting behaviour, and widespread micronutrient deficiencies, especially zinc (48%) and molybdenum (11%), which affect pulse crop production [57]. The soybean-chickpea cropping sequence is ideally

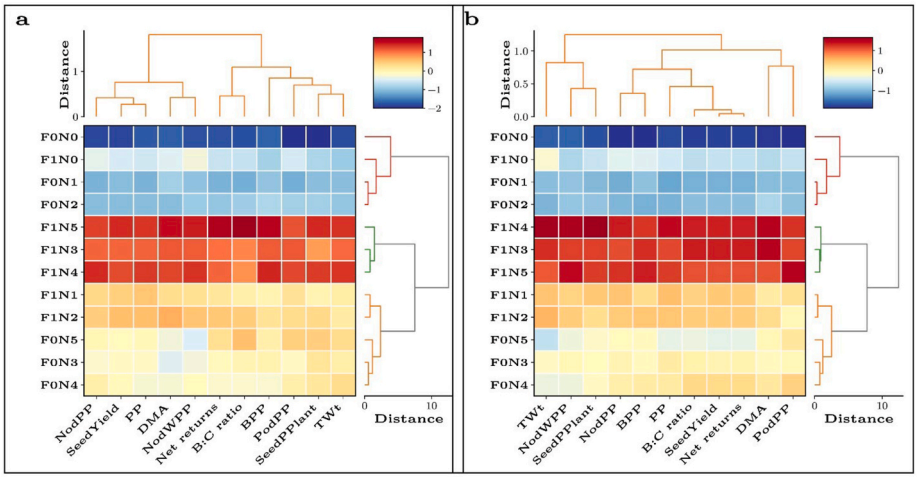


Fig. 7. Hierarchical clustering and heatmap analysis of growth, yield, and economic parameters in (a) soybean and (b) chickpea as influenced by FYM and micronutrient management. F0: without FYM application; F1: FYM @ 5 t ha^{-1}; N0: control; N1: $ZnSO_4$ @ 25 kg ha^{-1} soil application; N2: $ZnSO_4$ @ 12.5 kg ha^{-1} + Zn-solubilising bacteria; N3: ammonium molybdate @ 1 g kg^{-1} seed treatment + $ZnSO_4$ @ 25 kg ha^{-1} soil application; N4: ammonium molybdate @ 1 kg ha^{-1} + $ZnSO_4$ @ 25 kg ha^{-1}, both as soil application; N5: ammonium molybdate @ 1 g kg^{-1} seed treatment + $ZnSO_4$ @ 0.5% foliar spray. DMA: dry matter accumulation; BPP: branches per plant; NodPP: nodules per plant; NodWPP: nodule weight per plant; PodPP: pods per plant; SeedPPod: seeds per pod; SeedPP: seeds per plant; TWt: test weight; PP: protein productivity; Yield: grain yield.

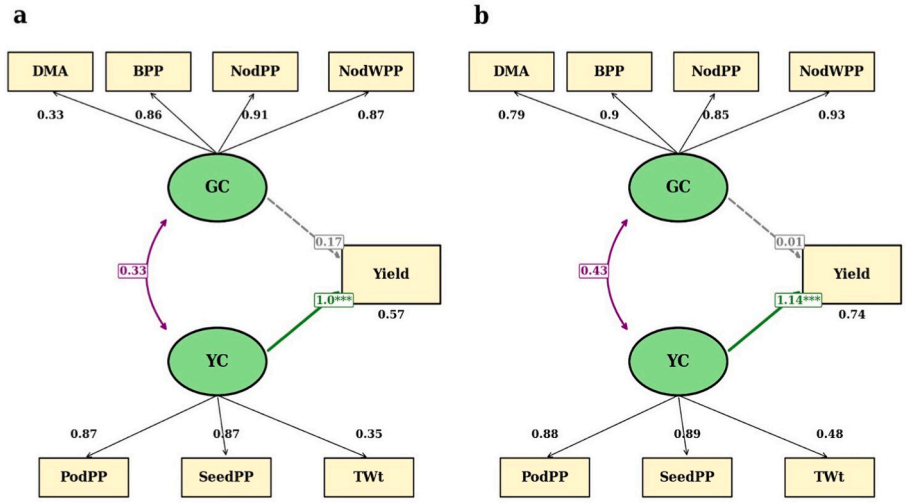


Fig. 8. Structural equation model (SEM) showing the relationships among growth characters (GC), yield characters (YC), and seed yield of soybean under (a) without FYM application (F0) and (b) with FYM application (F1). DMA: dry matter accumulation; BPP: branches per plant; NodPP: nodules per plant; NodWPP: nodule weight per plant; PodPP: pods per plant; SeedPP: seeds per plant; TWt: test weight; Yield: grain yield. Values on arrows indicate standardized path coefficients; purple curved arrows indicate covariance between GC and YC. ***p<0.001. Values adjacent to Yield box indicate variance explained (R^2).

suited for Vertisol because of complementary resource use, biological nitrogen fixation, and economic diversification [10]. However, realising the full yield potential requires strategic management of micronutrients and organic amendments. The present study assessed integrated nutrient management practices using farmyard manure along with micronutrients applied through different methods for the optimal productivity, profitability, and protein output of the soybean-chickpea system in rainfed Vertisols.

4.1. *Growth, yield attributes and yield of soybean*

Soybean showed significant responses to both farmyard manure and micronutrient management, with FYM application at 5 t ha^{-1} producing 25.8% higher seed yield compared to the control (Table 2). FYM improved all growth parameters, including dry matter accumulation (20.9%), nodule number (21.2%), nodule dry weight (27.7%), pods per plant (23.6%), and seeds per plant (16.2%), reflecting the dual role of FYM in Vertisols: improving soil physical properties and enhancing micronutrient availability in the alkaline soil environment [18,58]. In the present Vertisol (pH 8.21), inorganic Zn is predominantly immobilised through precipitation as $ZnCO_3$ and $Zn(OH)_2$, adsorption onto $CaCO_3$ surfaces, and co-precipitation with Fe and Mn oxides, which severely constrain the plant-available Zn pool [25,59]. At such alkaline pH, even the DTPA-extractable Zn pool (0.46 mg kg^{-1}, below the critical limit of 0.60 mg kg^{-1}; [23,43]) is only slowly replenished from the solid phase during periods of peak crop demand, and the addition of FYM-derived organic ligands accelerates this replenishment [42,60]. The organic acids released during FYM decomposition, particularly citric, oxalic, and fulvic acids, form soluble organo-Zn complexes that resist fixation and maintain Zn mobility in the rhizosphere through ligand exchange mechanisms [59]. Despite the variable effects of different organic fractions on Zn sorption reported in the literature [60],

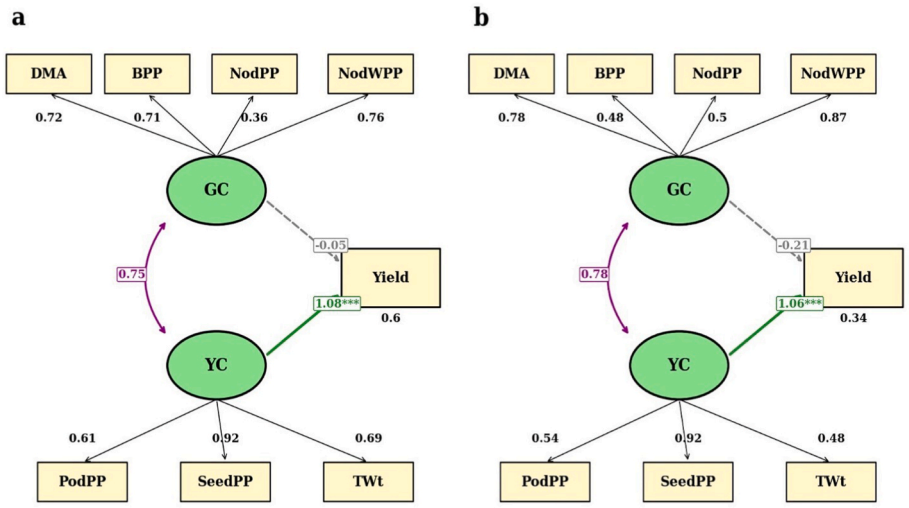


Fig. 9. Structural equation model (SEM) showing the relationships among growth characters (GC), yield characters (YC), and seed yield of chickpea under (a) without FYM application (F0) and (b) with FYM application (F1). DMA: dry matter accumulation; BPP: branches per plant; NodPP: nodules per plant; NodWPP: nodule weight per plant; PodPP: pods per plant; SeedPP: seeds per plant; TWt: test weight; Yield: grain yield. Values on arrows indicate standardized path coefficients; purple curved arrows indicate covariance between GC and YC. ***$p<0.001$. Values adjacent to Yield box indicate variance explained (R^2).

the net effect of FYM on Zn availability was clearly positive in the present study, as reflected in the consistent yield and growth improvements under F1 (Table 2). This organo-metallic complexation is the principal mechanism through which FYM enhances micronutrient bioavailability in calcareous soils [18,19].

The combined application of Mo and Zn (N3, N4, and N5) consistently outperformed single Zn application (N1, N2), with the highest soybean seed yield of 1393 kg ha^{-1} under N5 (37.1% over control, Table 2). This synergistic response reflects the complementary roles of these two micronutrients in legume nitrogen metabolism. Molybdenum is the catalytic centre of the Fe-Mo cofactor of nitrogenase, which drives biological nitrogen fixation in root nodules [32,33,35]. The combined Zn + Mo treatments increased nodule number by 25-30% and nodule dry weight by 28-32% over the control (Table 2), compared to only 12-13% and 13-16% under single Zn application (N1, N2), providing direct evidence that Mo addition substantially enhanced symbiotic nitrogen fixation. Molybdenum also functions as a cofactor of nitrate reductase, enabling assimilation of soil-derived nitrate [34]. Simultaneously, Zn is required for enzymes that channel the fixed and assimilated nitrogen toward protein synthesis, including carbonic anhydrase for photosynthetic carbon supply and RNA polymerase for seed protein accumulation [28,61]. Without adequate supply of both micronutrients, neither the nitrogen fixation pathway (Mo-dependent) nor the nitrogen utilisation pathway (Zn-dependent) operates at full capacity. This mechanistic interdependence is evident in the present data: combined treatments (N3-N5) produced 34-37% yield increases compared to 18-19% under single Zn (N1-N2), a response gap attributable to the additional Mo-driven nitrogen fixation benefit.

Among the micronutrient application methods, foliar Zn (N5) produced the highest soybean yield (1393 kg ha^{-1}), though statistically at par with N4 (1377 kg ha^{-1}) and N3 (1357 kg ha^{-1}). The numerical superiority of foliar application reflects two advantages: foliar-applied Zn bypasses soil fixation at alkaline pH and is directly absorbed through stomatal and cuticular pathways for rapid translocation to metabolically active tissues [29,62], and application at 30 and 60 DAS coincided with pre-flowering and early pod developmental stages when Zn demand is critical for reproductive organ development [24,27]. The micronutrient treatments (N3-N5) increased pods per plant by 39-46% and seeds per plant by 35-43% over the control (Table 2), consistent with the established role of Zn in pollen viability, ovule fertilisation, and seed set in grain legumes [26,28,31]. The short growth duration of soybean (97 days) and its shallower root system further favour timely foliar nutrient delivery [24]. However, this foliar advantage did not extend to the residual crop (Section 4.2), where soil-applied micronutrients provided greater carryover to chickpea, confirming hypothesis (iii) of the study.

4.2. Residual effect on growth, yield attributes and yield of chickpea

Chickpea received no direct application of farmyard manure or micronutrients, yet showed substantial yield benefits from residual effects of treatments applied to the preceding soybean crop. Residual FYM led to a 19.2% increase in chickpea seed yield and a 45.5% increase in nodule dry weight (Table 3), a proportionally greater nodulation response than the direct FYM effect observed in soybean (27.7%). This enhanced residual response can be attributed to the persistence of FYM-derived organic matter in Vertisols, which continues to improve soil structure, water retention, and cation exchange capacity over multiple seasons [18,20]. Progressive mineralisation of the remaining FYM releases nutrients (phosphorus, sulphur) and sustains populations of native rhizobia (*Mesorhizobium ciceri*), thereby creating a more favourable rhizosphere environment for chickpea nodulation [19,63–65].

Treatment N4 (soil-applied Mo + Zn) produced the highest residual chickpea seed yield of 2138 kg ha^{-1} (31.2% over control), followed by N3 (29.2%) and N5 (24.3%) (Table 3). This ranking reversal compared to soybean, where foliar Zn (N5) was numerically superior, directly confirms hypothesis (iii) of the present study: the application method optimal for the directly-fertilised crop differs from the method providing the greatest residual benefit. Soil-applied micronutrients create a residual nutrient pool in the soil matrix that persists beyond the first crop, whereas foliar-applied Zn is utilised entirely by the soybean canopy and contributes negligibly to the soil residual pool. The deeper and more adaptive root system of chickpea, which shifts root distribution to deeper soil layers under water deficit conditions [66], can effectively access these residual nutrients from soil layers not fully exploited by soybean, which concentrates over 95% of its roots in the surface 23 cm [66]. This finding has direct practical significance: for the soybean-chickpea system, soil application of both micronutrients (N4) maximises the combined return across both crops despite not being the highest-yielding treatment in soybean alone.

The importance of residual Mo for chickpea is evident from the nodulation data. Combined Mo + Zn treatments (N3-N5) increased chickpea nodule dry weight by 69-71% over the control, compared to 35-37% under single Zn application (N1-N2) (Table 3). This differential

nodulation response closely parallels the yield gap between combined and single treatments (24-31% vs 14% yield increase), confirming that Mo-driven enhancement of symbiotic nitrogen fixation is the primary mechanism underlying the residual yield benefit. Molybdenum serves dual enzymatic roles: as the catalytic centre of nitrogenase for atmospheric N2 fixation [32,33] and as a cofactor of nitrate reductase for soil nitrate assimilation [34,35]. Adequate residual Mo ensures efficient utilisation of both nitrogen sources, simultaneously driving grain filling and protein accumulation. The strong positive correlation between seed yield and protein productivity in chickpea ($r = 0.97$, $p < 0.001$; Fig. 5) reflects this tight coupling between yield formation and nitrogen metabolism [37].

4.3. System productivity, economic profitability, and practical implications

Treatment N4 resulted in the highest soybean equivalent yield of 4464 kg ha^{-1} (32.7% over control), with net returns of ₹94,332 ha^{-1} and a benefit-cost ratio of 3.52 (Table 4). The economic superiority of N4 over N5 reflects both the lower input cost of a single soil application compared to two foliar sprays and the substantially higher residual chickpea yield under N4 (2138 vs 2027 kg ha^{-1}, Table 3). System productivity under FYM (4439 kg ha^{-1}) was 21.2% higher than without FYM, and net returns were 33.4% higher (Table 4). The significant FYM × micronutrient interaction for system protein productivity ($p = 0.04$, Table 5) indicates that the protein enhancement from micronutrient application was greater under FYM-amended conditions, underscoring the synergistic nature of the INM approach [19,65,67].

From a practical standpoint, treatment N4 (soil-applied $ZnSO_4$ + ammonium molybdate with FYM) offers the most adoptable recommendation for smallholder farmers: all inputs are applied once at sowing, requiring no additional operations. For farmers with limited FYM access, combined Zn + Mo treatments alone provided 34-37% soybean and 24-31% chickpea yield increases over the control, demonstrating that micronutrient management is beneficial even without organic amendment.

4.4. Protein productivity and protein supply capacity

Treatment N4 recorded the highest system protein productivity of 1081 kg ha^{-1} (46.9% over control), comprising 574 kg ha^{-1} from soybean and 507 kg ha^{-1} from chickpea (Table 5). The positive relationship between protein content and seed yield evident in the regression analysis (Fig. 6) confirms that INM treatments enhanced both yield and grain protein concentration without the dilution effect commonly observed in cereals [68]. This absence of a yield-protein trade-off is attributable to Mo-driven enhancement of biological nitrogen fixation, which supplies additional nitrogen for both grain filling and protein synthesis simultaneously [32,37,69]. The protein supply capacity under N4 was 51 persons ha^{-1} year^{-1} (47% over control, Table 5). As described in the methodology, this is a theoretical comparative index based on the ICMR-recommended 58 g protein person^{-1} day^{-1} [54] and does not account for protein digestibility or end-use allocation, but provides a useful measure of cropping system performance for comparing INM strategies.

4.5. Yield determinants through structural equation modelling

Structural equation modelling partitioned the direct and indirect pathways through which growth and yield characters influence seed yield [55]. In both crops, yield characters (YC) were the decisive determinants ($\beta = 1.0–1.14$ in soybean, $\beta = 1.06–1.08$ in chickpea; $p < 0.001$), whereas growth characters (GC) showed non-significant direct effects (Figs. 8 and 9), indicating that INM treatments improve yield primarily through enhanced reproductive partitioning rather than vegetative growth. The GC-YC covariance was higher in chickpea (0.75–0.78) than in soybean (0.33–0.43), reflecting the tighter coupling between vegetative and reproductive phases in the determinate chickpea compared to indeterminate soybean. FYM application strengthened the soybean model (R^2 from 0.57 to 0.74) but reduced chickpea model fit (R^2 from 0.60 to 0.34), suggesting that FYM introduces additional unmeasured yield-determining factors in chickpea.

5. Conclusion

The three-year field evaluation confirmed that integrated application of farmyard manure at 5 t ha^{-1} with combined zinc and molybdenum significantly improved soybean productivity and provided substantial residual benefits to the subsequent chickpea. Soil application of $ZnSO_4$ @ 25 kg ha^{-1} and ammonium molybdate @ 1 kg ha^{-1} along with FYM @ 5 t ha^{-1} resulted in the highest system productivity, economic returns, and protein supply capacity. The application method optimal for soybean (ammonium molybdate @ 1 g kg^{-1} seed treatment + foliar application of $ZnSO_4$ @ 0.5%) differed from that providing the greatest residual benefit to chickpea (soil-applied zinc and molybdenum), confirming that system-level optimisation requires consideration of both the directly-fertilised and residual crops. Yield characters were identified through structural equation modelling as the decisive yield determinants, indicating that INM treatments improve productivity through reproductive partitioning rather than vegetative growth. Future research should integrate periodic soil and plant micronutrient monitoring to assess long-term accumulation risk and quantify residual nutrient availability. Evaluation of advanced formulations alongside conventional sources and assessment of environmental sustainability indicators would further strengthen integrated micronutrient management recommendations for legume-based cropping systems of rainfed Vertisols.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this revised manuscript, the authors used Claude Opus 4.6 (Anthropic) for language editing and structural reorganization of the manuscript text. Paperpal was used for reference formatting and management. The authors reviewed and edited all AI-generated suggestions and take full responsibility for the content, accuracy, and integrity of the published work.

CRediT authorship contribution statement

Ram Lal Jat: Conceptualization, Data curation, Investigation, Methodology, Supervision. **Lalit Kumar Rolaniya:** Formal analysis, Software, Visualization, Writing – original draft. **Chaitanya Prasad Nath:** Writing – original draft. **Chandra Sekhar Praharaj:** Investigation, Supervision. **Monika Punia:** Visualization, Writing – review & editing. **Raja Ram Choudhary:** Writing – review & editing. **Ram Swaroop Dadarwal:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jafr.2026.103080.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author upon reasonable request.

References

[1] B.R. Sharma, K.V. Rao, K.P.R. Vittal, Y.S. Ramakrishna, U. Amarasinghe, Estimating the potential of rainfed agriculture in India: prospects for water productivity improvements, Agric. Water Manag. 97 (2009) 23–30, https://doi.org/10.1016/j.agwat.2009.08.002.

[2] C. Srinivasa Rao, R. Lal, J.V.N.S. Prasad, K.A. Gopinath, R. Singh, V.S. Jakkula, K. L. Sahrawat, B. Venkateswarlu, A.K. Sikka, S.M. Virmani, Potential and challenges of rainfed farming in India, Adv. Agron. 133 (2015) 113–181, https://doi.org/10.1016/bs.agron.2015.05.004.

[3] T.P. Chibarabada, A.T. Modi, T. Mabhaudhi, Expounding the value of grain legumes in the semi- and arid tropics, Sustainability 9 (2017) 60, https://doi.org/10.3390/su9010060.

[4] M.A. Hanjra, M.E. Qureshi, Global water crisis and future food security in an era of climate change, Food Policy 35 (2010) 365–377, https://doi.org/10.1016/j.foodpol.2010.05.006.

[5] B.K. Rajashekhara Rao, K. Krishnappa, C. Srinivasarao, S.P. Wani, K.L. Sahrawat, G. Pardhasaradhi, Alleviation of multinutrient deficiency for productivity enhancement of rain-fed soybean and finger millet in the semi-arid region of India, Commun. Soil Sci. Plant Anal. 43 (2012) 1427–1435, https://doi.org/10.1080/00103624.2012.670344.

[6] K.H.M. Siddique, C. Johansen, N.C. Turner, M.-H. Jeuffroy, A. Hashem, D. Sakar, Y. Gan, S.S. Alghamdi, Innovations in agronomy for food legumes. A review, Agron. Sustain. Dev. 32 (2012) 45–64, https://doi.org/10.1007/s13593-011-0021-5.

[7] P.H. Graham, C.P. Vance, Legumes: importance and constraints to greater use, Plant Physiol. 131 (2003) 872–877, https://doi.org/10.1104/pp.017004.

[8] D.K. Agarwal, S.D. Billore, A.N. Sharma, B.U. Dupare, S.K. Srivastava, Soybean: introduction, improvement, and utilization in India—Problems and prospects, Agric. Res. 2 (2013) 293–300, https://doi.org/10.1007/s40003-013-0088-0.

[9] M. Imtiaz, R.S. Malhotra, S.S. Yadav, Genetic Adjustment to Changing Climates: Chickpea, 2011, pp. 251–268, https://doi.org/10.1002/9780470960929.ch18.

[10] N. Kumawat, M.L. Jadav, D.V. Bhagat, S.K. Choudhary, D. Chourey, Evaluation of different soybean based rainfed cropping sequences in vertisols of Madhya Pradesh, Indian J. Dryland Agric. Res. Dev. 37 (2022) 15–19, https://doi.org/10.5958/2231-6701.2022.00004.5.

[11] K. Ramteke, V.V. Gabhane, M.S. Bhavsar, A. Devatale, R.S. Patode, M.M. Ganvir, B. A. Sonune, Impact of conservation tillage on soil fertility, productivity and nutrient uptake by chickpea in soybean-chickpea sequence on vertisols, Int. J. Plant Soil Sci. 36 (2024) 423–429, https://doi.org/10.9734/ijpss/2024/v36i94991.

[12] S. Swaminathan, R. Hemalatha, A. Pandey, N.J. Kassebaum, A. Laxmaiah, T. Longvah, et al., The burden of child and maternal malnutrition and trends in its indicators in the states of India: the global burden of disease Study 1990–2017, Lancet child Adolesc, Health 3 (2019) 855–870, https://doi.org/10.1016/S2352-4642(19)30273-1.

[13] R.S. Bhatnagar, O.I. Padilla-Zakour, Plant-based dietary practices and socioeconomic factors that influence anemia in India, Nutrients 13 (2021) 3538, https://doi.org/10.3390/nu13103538.

[14] B.U. Dupare, Improved Technologies and Recommendations for Maximizing Soybean Productivity, Extension Bulletin No. 18, ICAR-Indian Institute of Soybean Research, 2023, p. 74. https://icar-nsri.res.in/pdfdoc/ExtensionBulletin2023E_2.pdf. (Accessed 2 June 2026).

[15] P. Ramesh, N. Raten Panwar, A. Bahadur Singh, S. Ramana, A. Subba Rao, Impact of organic-manure combinations on the productivity and soil quality in different cropping systems in Central India, J. Plant Nutr. Soil Sci. 172 (2009) 577–585, https://doi.org/10.1002/jpln.200700281.

[16] M. Diacono, F. Montemurro, Long-term effects of organic amendments on soil fertility, in: E. Lichtfouse, M. Hamelin, M. Navarrete, P. Debaeke (Eds.), Sustainable Agriculture Volume 2, Springer Netherlands, Dordrecht, 2011, pp. 761–786, https://doi.org/10.1007/978-94-007-0394-0_34.

[17] D.K. Inwati, B.K. Dixit, R. Thakur, B.S. Dwivedi, A. Sharma, V. Singh, Effect of long term fertilizer and FYM application on soil fertility of a vertisol under soybean-wheat cropping system, J. Exp. Agric. Int. 44 (2022) 179–184, https://doi.org/10.9734/jeai/2022/v44i1030891.

[18] J. Bangre, N.K. Sinha, M. Mohanty, S. Jayaraman, R.N. Sahoo, A.K. Dwivedi, et al., Long-term impact of inorganic fertilizers and farmyard manure on soil quality and productivity in subtropical vertisols under a soybean-wheat cropping system, Land Degrad. Dev. 35 (2024) 4257–4270, https://doi.org/10.1002/ldr.5220.

[19] R.L. Jat, C.S. Praharaj, Impact of zinc and molybdenum with manure in soybean-chickpea system in Vertisols of Central India, J. Food Legumes 31 (2018) 147–153. https://pub.isprd.in/index.php/jfl/article/view/428/270. (Accessed 25 December 2025).

[20] R.L. Jat, L.K. Rolaniya, M. Punia, R.R. Choudhary, Residual effect of organic and inorganic fertilizers on growth indices of chickpea under soybean-chickpea system in vertisol of Central India, Int. J. Plant Soil Sci. 35 (2023) 75–83, https://doi.org/10.9734/ijpss/2023/v35i244299.

[21] M.V. Singh, Micronutrient nutritional problems in soils of India and improvement for human and animal health, Indian J. Fertil. 5 (2009) 11–26. https://www.cabidigitallibrary.org/doi/full/10.5555/20093121241. (Accessed 2 June 2026).

[22] M. Chakraborty, D. Chakraborty, P. Mondal, R. Paul, Micronutrient deficiency stress in soils of India: tackling it to alleviate hidden hunger, in: A. Rakshit, S. K. Singh, P.C. Abhilash, A. Biswas (Eds.), Soil Science: Fundamentals to Recent Advances, Springer Singapore, Singapore, 2021, pp. 801–821, https://doi.org/10.1007/978-981-16-0917-6_40.

[23] J.S. Khokhar, M.R. Broadley, E.L. Ander, Soil zinc surveillance frameworks can inform human nutrition studies: opportunities in India, Front. Soil Sci. 4 (2024) 1421652, https://doi.org/10.3389/fsoil.2024.1421652.

[24] NAAS, Sustaining Soybean Productivity and Production in India, National Academy of Agricultural Sciences, New Delhi, 2017, p. 52. https://naas.org.in/spapers/Strategy%20Paper%20No.%204.pdf. (Accessed 2 June 2026).

[25] Ch Srinivasarao, K.P.R. Vittal, Emerging nutrient deficiencies in different soil types under rainfed production systems of India, Indian J. Fertil. 3 (2007) 37–44. https://www.researchgate.net/publication/284049162. (Accessed 3 June 2026).

[26] H.R. Khan, G.K. McDonald, Z. Rengel, Zinc fertilization and water stress affects plant water relations, stomatal conductance and osmotic adjustment in chickpea (*Cicer arientinum* L.), Plant Soil 267 (2004) 271–284, https://doi.org/10.1007/s11104-005-0120-7.

[27] S. Kachhi, G.S. Tagore, S. Yadav, Effect of frequency and rate of zinc application on nodulation, chlorophyll content, yield, and quality of soybean grown in a vertisol of Jabalpur, Madhya Pradesh, India, Int. J. Plant Soil Sci. 35 (2023) 937–947, https://doi.org/10.9734/ijpss/2023/v35i214064.

[28] G. Hacisalihoglu, Zinc (Zn): the last nutrient in the alphabet and shedding light on Zn efficiency for the future of crop production under suboptimal Zn, Plants 9 (2020) 1471, https://doi.org/10.3390/plants9111471.

[29] S.S. Dhaliwal, V. Sharma, A.K. Shukla, V. Verma, M. Kaur, Y.S. Shivay, et al., Biofortification—A frontier novel approach to enrich micronutrients in field crops to encounter the nutritional security, Molecules 27 (2022) 1340, https://doi.org/10.3390/molecules27041340.

[30] D.B. Ajjannavar, V.G. Yadahalli, S. Menasinakai, M. Awaradi, V.D. Malakai, Combined application of micronutrients and biofertilizers on the yield and quality of chickpea (*Cicer arietinum* L.) in vertisols, Int. J. Plant Soil Sci. 37 (2025) 552–561, https://doi.org/10.9734/ijpss/2025/v37i85655.

[31] S.K. Ahirwar, A. Kumar, R. Bhargava, R. Patel, D. Dwarka, O. Singh, et al., The effect of various doses of zinc on the growth and yield of soybean crop, *Glycine max* L, Int. J. Plant Soil Sci. 35 (2023) 308–312, https://doi.org/10.9734/ijpss/2023/v35i244321.

[32] M. Imran, X. Sun, S. Hussain, U. Ali, M.S. Rana, F. Rasul, et al., Molybdenum-induced effects on nitrogen metabolism enzymes and elemental profile of winter wheat (*Triticum aestivum* L.) under different nitrogen sources, Int. J. Mol. Sci. 20 (2019) 3009, https://doi.org/10.3390/ijms20123009.

[33] R.R. Mendel, Molybdoenzymes and molybdenum cofactor in plants, J. Exp. Bot. 53 (2002) 1689–1698, https://doi.org/10.1093/jxb/erf038.

[34] M. Yu, C.-X. Hu, X.-C. Sun, Y.-H. Wang, Influences of Mo on nitrate reductase, glutamine synthetase and nitrogen accumulation and utilization in Mo-efficient and Mo-inefficient winter wheat cultivars, Agric. Sci. China 9 (2010) 355–361, https://doi.org/10.1016/S1671-2927(09)60104-8.

[35] K. Béla, E. Bodi, S. Várallyay, S.A. Mukarram, A.E.M. Ahmed, Enhancing plant growth and yield quality through molybdenum application: trends, future and directions, Plants 15 (2026) 585, https://doi.org/10.3390/plants15040585.

[36] Rajmata Vijayaraje Scindia Krishi Vishwavidyalaya (RVSKVV), Significant achievements, RVSKVV Gwalior official website, https://www.rvskvv.net/index.php/research/significant-achievements (accessed 1 June 2026).

[37] M.S. Venkatesh, K.K. Hazra, P.K. Ghosh, S.S. Singh, S.K. Chaturvedi, Response of chickpea (*Cicer arietinum*) to molybdenum in moderately-alkaline soil, Indian J. Agric. Sci. 90 (2020) 58–63, https://doi.org/10.56093/ijas.v90i1.98531.

[38] A. Suryawanshi, G.S. Tagore, P.S. Kulhare, G.D. Sharma, Effect of molybdenum and inoculants on symbiotic traits, nodule leghaemoglobin and yield of soybean in a vertisol, Ecol. Environ. Conserv. 22 (2016) 335–338. https://www.envirobiotechjournals.com/issues/article_abstract.php?aid=6819&iid=209&jid=3. (Accessed 1 June 2026).

[39] D. Vijaya Lakshmi, T. Prabhakar Reddy, K. Sridhar, M. Parimala Kumar, Effect of molybdenum seed treatment on yield and economics of chickpea (*Cicer arietinum* L.), Int. J. Agric. Ext. Soc. Dev. 9 (2026) 448–450, https://doi.org/10.33545/26180723.2026.v9.i3g.3418.

[40] C. Deo, M.L. Kothari, Effect of modes and levels of molybdenum application on grain yield protein content and nodulation of chickpea grown on loamy sand soil, Commun. Soil Sci. Plant Anal. 33 (2002) 2905–2915, https://doi.org/10.1081/CSS-120014490.

[41] B. Rama Devi, V. Singh, D. Tiwari, M. Sudhakar, M. Rakesh, Influence of organic manures and molybdenum on growth and yield of groundnut (*Arachis hypogaea* L.), Biol. Forum Int. J. 13 (2021) 402–406. https://www.researchtrend.net/bfij/influence-of-organic-manures-and-molybdenum-on-growth-and-yield-of-groundnut-arachis-hypogaea-l-3068. (Accessed 2 June 2026).

[42] S. Singh, G.S. Tagore, S. Singh, S. Yadav, Effect of zinc-enriched organic manures on zinc composition in wheat and zinc pools in a vertisols of Central India, AATCC

Rev. 13 (2025) 430–436, https://doi.org/10.21276/AATCCReview.2025.13.04.430.
[43] T. Prabhakar Reddy, D. Vijaya Lakshmi, J. Kamalakar, Ch Sambasiva Rao, Critical levels of micro and secondary nutrients in soils and crops for optimum plant nutrition, Int. J. Sci. Res. 6 (2017) 594–595. https://www.researchgate.net/publication/348428664. (Accessed 2 June 2026).
[44] M. Kumar, M. Kumar, R.K. Singh, A.M. Waghaye, V.B. Babu, R.D. Randhe, Trend detection and prediction of rainfall using logistic machine learning and deep learning models in Bhopal region of Central India, J. Water Clim. Change 16 (2025) 1055–1072, https://doi.org/10.2166/wcc.2025.662.
[45] L.A. Richards, Diagnosis and improvement of saline and alkali soils, Soil Sci. 78 (1954) 154, https://doi.org/10.1097/00010694-195408000-00012.
[46] A. Walkley, I.A. Black, An examination of the Degtjareff method for determining soil organic matter and a proposed modification of the chromic acid titration method, Soil Sci. 37 (1934) 29–38, https://doi.org/10.1097/00010694-193401000-00003.
[47] B.V. Subbiah, G.L. Asija, A rapid procedure for the estimation of available nitrogen in soils, Curr. Sci. 25 (1956) 259–260. https://www.cabidigitallibrary.org/doi/full/10.5555/19571900070. (Accessed 24 December 2025).
[48] S.R. Olsen, C.V. Cole, F.S. Watanabe, L.A. Dean, Estimation of available phosphorus in soils by extraction with sodium bicarbonate, USDA Circular No. 939 (1954). Washington, DC, https://www.cabidigitallibrary.org/doi/full/10.5555/19541901890. (Accessed 24 December 2025).
[49] M.L. Jackson, Soil Chemical Analysis, Prentice Hall, Inc., Englewood Cliffs, NJ, 1958. https://archive.org/details/soilchemicalanal030843mbp. (Accessed 10 June 2026).
[50] W.L. Lindsay, W.A. Norvell, Development of a DTPA soil test for zinc, iron, manganese, and copper, Soil Sci. Soc. Am. J. 42 (1978) 421–428, https://doi.org/10.2136/sssaj1978.03615995004200030009x.
[51] S.C. Gupta, S. Gangwar, Effect of molybdenum, iron and microbial inoculants on symbiotic traits, nutrient uptake and yield of chickpea, J. Food Legumes 25 (2012) 45–49. https://pub.isprd.in/index.php/jfl/article/view/1212/803. (Accessed 3 June 2026).
[52] P. Somasegaran, H.J. Hoben, Handbook for Rhizobia: Methods in legume-Rhizobium Technology, Springer, New York, NY, 1994, https://doi.org/10.1007/978-1-4613-8375-8.
[53] AOAC, Official Method 979.09, protein in grains, in: G.W. Latimer Jr. (Ed.), Official Methods of Analysis of AOAC INTERNATIONAL, twenty-second ed., AOAC Publications, New York, 2023 https://doi.org/10.1093/9780197610145.003.2973. (Accessed 10 June 2026).
[54] Indian Council of Medical Research (ICMR), Nutrient Requirements and Recommended Dietary Allowances for Indians: a Report of the Expert Group of the Indian Council of Medical Research, National Institute of Nutrition, Hyderabad, India, 2010. https://enacnetwork.com/files/pdf/ICMR_RDA_BOOK_2010.pdf. (Accessed 2 June 2026).
[55] Y. Rosseel, Lavaan: an R package for structural equation modeling, J. Stat. Software 48 (2012), https://doi.org/10.18637/jss.v048.i02.
[56] D.K. Pal, P. Tiwary, Pedogenic processes derived plant available water capacity (PAWC) for rain-fed deep-rooted crops in Indian vertisols: a pragmatic method, Agropedology 32 (2019) 94–103, https://doi.org/10.47114/j.agroped.2022.jun6.
[57] M.V. Singh, Micronutrient deficiencies in crops and soils in India, in: B.J. Alloway (Ed.), Micronutrient Deficiencies in Global Crop Production, Springer Netherlands, Dordrecht, 2008, pp. 93–125, https://doi.org/10.1007/978-1-4020-6860-7_4.
[58] M. Diacono, F. Montemurro, Long-term effects of organic amendments on soil fertility. A review, Agron. Sustain. Dev. 30 (2010) 401–422, https://doi.org/10.1051/agro/2009040.
[59] B. Sadeghzadeh, Z. Rengel, Zinc in soils and crop nutrition, in: M.J. Hawkesford, P. Barraclough (Eds.), The Molecular and Physiological Basis of Nutrient Use Efficiency in Crops, 2011, pp. 335–375, https://doi.org/10.1002/9780470960707.ch16.
[60] S.S. Dhaliwal, R.K. Naresh, A. Mandal, R. Singh, M.K. Dhaliwal, Dynamics and transformations of micronutrients in agricultural soils as influenced by organic matter build-up: a review, Environ. Sustain. Indic. 1–2 (2019) 100007, https://doi.org/10.1016/j.indic.2019.100007.
[61] M.R. Broadley, P.J. White, J.P. Hammond, I. Zelko, A. Lux, Zinc in plants, New Phytol. 173 (2007) 677–702, https://doi.org/10.1111/j.1469-8137.2007.01996.x.
[62] V. Fernández, P.H. Brown, From plant surface to plant metabolism: the uncertain fate of foliar-applied nutrients, Front. Plant Sci. 4 (2013) 289, https://doi.org/10.3389/fpls.2013.00289.
[63] I. Benjelloun, I. Thami Alami, M. El Khadir, A. Douira, S.M. Udupa, Co-inoculation of *Mesorhizobium ciceri* with either *Bacillus* sp. or *Enterobacter aerogenes* on chickpea improves growth and productivity in phosphate-deficient soils in dry areas of a Mediterranean region, Plants 10 (2021) 571, https://doi.org/10.3390/plants10030571.
[64] D. Gite, S.S. Patil, H. Gite, A.G. Bag, A. Dambale, Influence of residual integrated nutrient management practices on chickpea yield and nutrient dynamics in maize-chickpea cropping sequence, Indian J. Agric. Res. 59 (2025) 731–736, https://doi.org/10.18805/IJARe.A-6342.
[65] B. Meena, A.K. Biswas, R.H. Wanajri, R.S. Chaudhary, Integrated plant nutrient supply modules for enhancing crop growth, productivity and nutrient balance in maize-chickpea cropping sequence, Indian J. Agron. 69 (2024) 151–157, https://doi.org/10.59797/ija.v69i2.5500.
[66] J.G. Benjamin, D.C. Nielsen, Water deficit effects on root distribution of soybean, field pea and chickpea, Field Crops Res. 97 (2005) 248–253, https://doi.org/10.1016/j.fcr.2005.10.005.
[67] C.K. Dotaniya, B.L. Lakaria, Y. Sharma, B.P. Meena, S.B. Aher, A.O. Shirale, et al., Performance of chickpea (*Cicer arietinum* L.) in maize-chickpea sequence under various integrated nutrient modules in a vertisol of Central India, PLoS One 17 (2022) e0262652, https://doi.org/10.1371/journal.pone.0262652.
[68] M.J. Hawkesford, Reducing the reliance on nitrogen fertilizer for wheat production, J. Cereal. Sci. 59 (2013) 276–283, https://doi.org/10.1016/j.jcs.2013.12.001.
[69] E. Kebede, Contribution, utilization, and improvement of legumes-driven biological nitrogen fixation in agricultural systems, Front. Sustain. Food Syst. 5 (2021) 767998, https://doi.org/10.3389/fsufs.2021.767998.